

Lesson 01 Preparation: Squares and Cubes

Get ready for *Square and Cube Roots*

Name: _____

Class: MYP 8A

Date: _____

Purpose and time

Why? Next lesson you will learn to *undo* squaring and cubing. This sheet makes sure squares and cubes are fresh in your mind.

Time: about 10–15 minutes. Work on your own. No calculator is needed. Bring this sheet to class.

If you get stuck

1. Read the **Remember** box and the **Worked Example** again.
2. Write the multiplication out in full, for example $4^3 = 4 \times 4 \times 4$, and multiply two numbers at a time.
3. Use the list of squares and cubes in the Remember box.
4. Still stuck? Tick “I need help” at the end and write your question. That is useful for your teacher.

Part 1: Retrieval

what do you remember?

P1. Calculate.

(a) $6^2 =$ _____ (b) $9^2 =$ _____ (c) $12^2 =$ _____ (d) $2^3 =$ _____ (e) $4^3 =$ _____

P2. Write 5^3 as a multiplication. Then calculate it.

$5^3 =$ _____ $=$ _____

P3. Circle the numbers that are perfect squares: 20 25 36 50 81 90

P4. Is 64 a perfect square, a perfect cube, both, or neither? Explain.

Remember

Square: $n^2 = n \times n$ e.g. $7^2 = 7 \times 7 = 49$

A **perfect square** is a whole number squared.

Cube: $n^3 = n \times n \times n$ e.g. $2^3 = 2 \times 2 \times 2 = 8$

A **perfect cube** is a whole number cubed.

Careful: 3^2 means $3 \times 3 = 9$, *not* 3×2 .

A negative number squared is positive: $(-3)^2 = 9$.

n	1	2	3	4	5	6	7	8	9	10	11	12
n^2	1	4	9	16	25	36	49	64	81	100	121	144
n^3	1	8	27	64	125	216	343	512	729	1000		

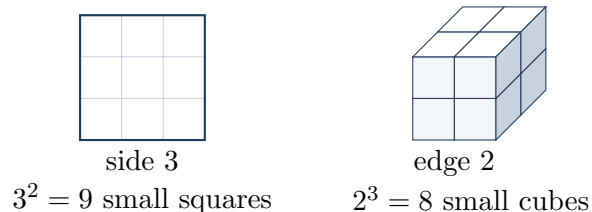
Part 2: Worked example

read each step

Worked Example: calculate 3^3 and $(-2)^3$

Step	Why
$3^3 = 3 \times 3 \times 3$	The small 3 tells us to use <i>three</i> 3s, multiplied.
$= 9 \times 3$	Multiply two at a time: $3 \times 3 = 9$.
$= 27$	Then $9 \times 3 = 27$. <i>Not</i> 9 and <i>not</i> $3 \times 3 = 9$.
$(-2)^3 = (-2) \times (-2) \times (-2)$	Three copies of -2 .
$= 4 \times (-2)$	negative $\times$ negative = positive, so $(-2) \times (-2) = 4$.
$= -8$	positive $\times$ negative = negative.
Check:	A cube of a negative number is negative. A square of a negative number is positive: $(-2)^2 = 4$.

A picture helps. Squaring gives the *area* of a square. Cubing gives the *volume* of a cube.



Part 3: Try it

show your working

P5. Cover the Remember box. Complete the table from memory, then uncover it and check.

n	1	2	3	4	5	6	7	8	9	10	11	12
n^2												
n^3												

(cubes up to 5^3 only)

P6. Calculate $(-5)^2$.

P7. Calculate $(-3)^3$.

P8. (a) A square has side length 8 cm. Find its area. (b) A cube has edge length 3 cm. Find its volume.

Part 4: Predict

a bridge to next lesson

P9. (a) A square has area 36 cm^2 . What is its side length? How did you work it out?
(b) Another square has area 40 cm^2 . Can you find its side length exactly? Between which two whole numbers must it lie? Why?

Self-check: tick one box for each

	I can do this	I need help
I can square a number, including a negative number.	☐	☐
I can cube a number, including a negative number.	☐	☐
I know the perfect squares up to 144 and perfect cubes up to 125.	☐	☐

My question for class: